

Md Samshad Rahman (MACS | M.Sc. Stats | B.Sc. CS)

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Professional summary:

Data Analyst with dual Master's degrees in Applied Computer Science and Applied Statistics and over 4 years of experience turning raw operational data into dashboards, anomaly detection, reporting, actionable insight for leadership. Background building ETL pipelines, running statistical validation, and tracking KPIs across utility billing, procurement, and retail operations. Comfortable working across SQL, Python, Machine Learning/AI, LLMs, and BI tools to move from a raw dataset to a decision optimally.

Core competencies:

Engineering Practices:	Agile Methodologies, Code Quality, Refactoring, Unit Testing, Integration Testing, System Reliability, Software Security, PyTest, TDD (Test-Driven Development), Automated Testing.
Analytics & Reporting:	Google Analytics, Dash/Plotly, Looker Studio, Tableau, Advanced Excel, KPI & Performance Metrics, Process Mapping, Business Process Re-engineering, Anomaly Detection, Data Validation & QA, Hypothesis-Driven Analysis.
Collaboration & Delivery:	Requirements Gathering, Stakeholder Relations, Technical Documentation, Agile/SDLC, Technical Training and Advisory.
DevOps & Tools:	Docker, Git, CI/CD (GitHub Actions), AWS (EC2, S3, Lambda), GCP, Jira, Confluence, CloudFormation, Kubernetes, Infrastructure as Code (IaC), Microservices Architecture, Pydantic, OpenAPI, Swagger.
Backend & APIs:	Python, SQL, JavaScript, FastAPI, Flask, REST APIs, PostgreSQL, MySQL, Redis, Celery, Taskiq.

Professional Experience:

Dalhousie University, Halifax, Nova Scotia

Data Specialist & Graduate Researcher (*Persuasive Computing Lab*)

[September 2024 – April 2025]

- Cut public procurement research time by 40% by replacing a legacy Nova Scotia government reporting workflow with automated Python ETL pipelines, standardizing 10 years of raw records and building an interactive Dash tool that removed 35% of dataset noise.
- Audited and extracted data from semi-structured contract and procurement text into a normalized PostgreSQL database, cutting manual review time by approximately 40%.
- Fine-tuned and deployed a 3B-parameter LLM on GPU infrastructure, balancing speed, accuracy, and cost for a production research tool.
- Built a multi-step reasoning agent pipeline (LangChain) and integrated a third-party NLP API to extend the tool's analytical capabilities.

Systems Analyst (*Data Integrity & Billing*), Munshi Engineers, Bangladesh

[February 2021 – April 2023]

- Built automated SQL and Python validation scripts across two national utility systems (DPDC, BPDB) that flagged consumption anomalies before monthly invoice runs, holding data integrity at 99.99% across high-volume billing records.
- Turned raw operational logs and transaction data into automated SQL and Power BI reports, leading a 4-person team through the process and helping leadership spot and stop revenue leakage.
- Replaced a resource-heavy prototype with an event-driven AWS Lambda pipeline, cutting infrastructure cost by approximately 50% while lifting processing throughput by over 30%.
- Wrote data dictionaries and validation protocols for national infrastructure datasets, keeping audits ready and data consistent across teams.

Python Developer, Smartbytes Ltd. Bangladesh

[February 2022 – September 2022]

- Built an NLP ingestion pipeline (spaCy, LinearSVC) that parsed 15,000+ unstructured documents into normalized PostgreSQL schemas, lifting matching accuracy from 65% to 92%.
- Standardized data from 5+ external providers into one normalized MySQL warehouse, resolving schema conflicts across ~50,000 monthly records with zero downtime.
- Automated the daily revenue reconciliation workflow, cutting manual consolidation time by over 80% (2 hours down to 15 minutes).
- Deployed the parsing engine as a FastAPI microservice with auto-generated documentation, cutting integration overhead for the frontend team.

Other Experience:

Microsoft Senior Advisor, Best Buy Canada Ltd. Halifax, NS

[October 2023 – Present]

- Analyzed weekly operational data to track KPI trends, working with team members and management to spot patterns and streamline technical onboarding and customer troubleshooting processes.

Education:

- **Master of Applied Computer Science** [September 2023 – May 2025]
Dalhousie University, Halifax, NS
- **Master's in Applied Statistics and Data Science** [November 2019 – April 2022]
Jahangirnagar University, Bangladesh
- **BSc. in Computer Science and Software Engineering** [January 2015 – January 2020]
American International University-Bangladesh